

Report 2026-09-24. Campaign run 2: a language model as one MATSim replanning strategy What was achieved, what was not, and what the model actually decides

Giulio Iannelli, MATSim AI Lab.

Repository `matrim-ai-lab`, branch `migrate/contrib-llm`, code at commit `a916897`

Run of 23 September 2026, 16:30–19:36. Report written 24 September 2026, revised 25 September 2026 (transfer-count bug, slower-mode analysis). Series: persona-emergence reports, no. 1.

Summary

MATSim simulates a city day after day. Each synthetic traveller keeps a few daily plans, executes one, gets a score for it and occasionally changes it with a rule-based strategy until the population settles. In this run, for 200 of 8,460 travellers, that strategy was a local 27-billion-parameter language model (LLM) that reads the day in the first person and decides per trip which mode to use. MATSim then scores the result like any other plan. The run: 25 iterations, 10 LLM queries per iteration, 3 h 06 min.

Achieved	The protocol is reliable and fast enough: 250 of 250 queries gave a valid, physically checked decision in one model round, 39 s median. The panel and trigger logic behaved as designed. The population stayed at rest, so the LLM’s effect is readable per agent. The reasoning is in the first person and grounded in the stated traits in 78 % of conversations.
Not achieved	No behavioural result. The model’s decision looks like a near-deterministic rule (a car owner drives, everyone else keeps the bus), but the rule is confounded by an input bug: the context reported two transfers for a direct bus. The decision findings must be re-measured after the fix. No learning: 20,799 experience facts were written to the vector store and none reached a prompt. MATSim’s scoring neither confirmed nor rejected the switches (car and bus score alike). No effect on the population, by design.
Issues	Transfers were over-reported in every bus option (direct bus shown as 2 transfers, one change as 4), and “transfers” was the reason cited most for switching to car. A context bug in the first attempt silently forbade every car switch (fixed, run repeated). A colleague’s job on the shared GPU forced a model reload in 117 of 250 queries (25 % of LLM time). The analysis suite does not understand the new decision format. Departure-time changes were never used.

1 Setting and question

Every MATSim agent owns up to six plans (a chain of activities with trips between them). Each iteration one plan per agent is executed on the network, scored (time at activities rewarded, travel time and cost penalised) and a fraction of agents replan: a copy of a plan is modified by a strategy, the others keep their best-scoring plan. The population relaxes to a user equilibrium after some tens of iterations.

The question is whether an LLM can be one of these strategies for a subset of agents, so that mode decisions are made by a persona with stated age, job and car access, while MATSim keeps the final word through scoring. Before any behavioural question can be asked, two things must hold: every query must yield a plan MATSim can run, and a run with hundreds of queries must fit in an afternoon. This run measures both, and then reads what the model decided and why.

2 Method

Ground. Sioux Falls (car, bus, walk), a 10 % population sample (8,460 agents) with capacities scaled to 0.1, run for 100 rule-based iterations beforehand: mean executed score $18.76 \rightarrow 20.16$, mode shares car/bus/walk $78/19/3 \% \rightarrow 61/28/11 \%$, no agent stuck. Every LLM run starts from these plans, so the population is at rest and any drift is caused by the LLM.

Panel and trigger. A seeded panel of 200 agents is tagged. Each iteration at most 10 of them go to the LLM; the rest of the panel and the 8,260 other agents replan with MATSim’s default strategies. Priority: stuck agents, then never-reviewed agents, then the 20 % of the panel with the worst drop of executed score since their last review. With 200 agents and 10 per iteration the first 20 iterations review everyone once; from iteration 21 the score-drop trigger drives selection.

One query, one round. Earlier runs let the model call tools over 6–9 rounds per agent, 375 s each. Here the facts are pre-computed into the first message: the activity chain; which vehicles the agent owns and where each is parked in the morning; for every trip the travel time, distance and transfers by car, bus and walk from MATSim’s router on today’s congested network. The model answers with one tool call, `decide_trips`: trips to change with a new mode and optional departure shift (empty list = keep the day). The decision is checked (trip

exists, mode known, the vehicle is with the traveller along the chain of trips) and MATSim routes the new legs. Prompt 2,673 tokens median; answer 745 tokens median, nearly all of it the reasoning trace.

Model and prompt. qwen3.6:27b via Ollama on a shared GPU server (one request at a time), temperature 0.3, context 12,288 tokens, hard cap 3,072 output tokens, model kept resident 10 min. The system prompt puts the model in the first person (“You are a 53-year-old man. You work. You have a car you can use whenever you want.”), states MATSim’s physical rules, and asks for short reasoning.

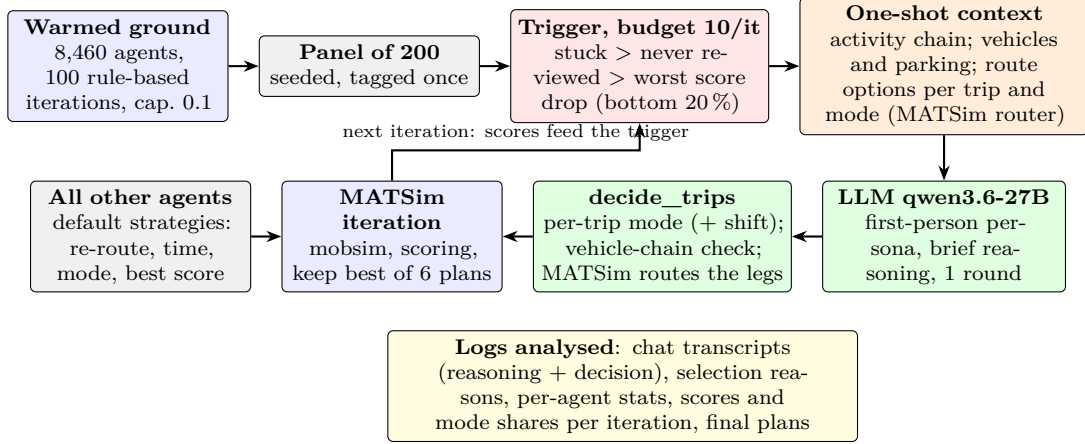


Figure 1: The loop. Ten panel agents per iteration are decided by the LLM; everyone else replans with MATSim’s rules. Scoring is the same for all plans, so an LLM plan survives only if it scores well. Nothing about the outcome is fed back to the persona.

3 What was achieved

3.1 A reliable and fast protocol

queries	250	decisions applied by MATSim	250
model rounds per query	1 (all)	retries / verification failures / cap hits	0 / 0 / 0
wall time per query, median (max)	39 s (105 s)	model resident / after a reload	32 s / 52 s
LLM time: generation / prompt / load	66 / 8 / 25 %	generation speed	26 tokens/s
run wall time	186 min	iteration, median (min–max)	6.4 min (5.4–9.7)

Table 1: Protocol and timing.

Every query ended with a valid decision in the first round; no agent was re-asked and no decision failed the chain check (Table 1). Query time is bimodal (Fig. 2b): 32 s with the model resident, 52 s after a reload. The reloads are not ours: a colleague ran the same model with a smaller context during iterations 1–3 and 17–25, and Ollama reloads the 17 GB weights on every switch (Fig. 3). A reload-free iteration takes 5.5 min, of which 5.3 min are the ten queries and 30 s is MATSim. Compared with the tool-calling protocol of the previous runs (375 s per agent, 6–9 rounds) this is a tenfold gain, obtained by pre-computing the facts and asking for a decision instead of a full plan.

3.2 The panel logic works and the ground stays at rest

Iterations 1–20 reviewed all 200 panel agents once; from iteration 21 the trigger selected 50 agents by score drop (median drop 0.22 utils, maximum 4.25), so 39 agents were queried twice or more (Fig. 6c). Mean executed score moved from 20.157 to 20.166 over the run, mode shares by under one point, and no agent was stuck at the end (Fig. 4). Ten decisions per iteration among 8,460 agents are invisible at population level. That is what a controlled instrument needs, and it is also why this run could not show behaviour (Section 5).

3.3 A persona voice survives the short-reasoning instruction

The traces speak in the first person (median 3.4 first-person markers per 100 words) and the detector counts the persona as “emerged” (first-person voice plus at least two grounded traits) in 195 of 250 conversations (Fig. 5). Age is cited in 241 traces, car ownership in 197, employment in 52. No repetition loops, no contradiction between a stated trait and a decision (for example a carless agent choosing car).

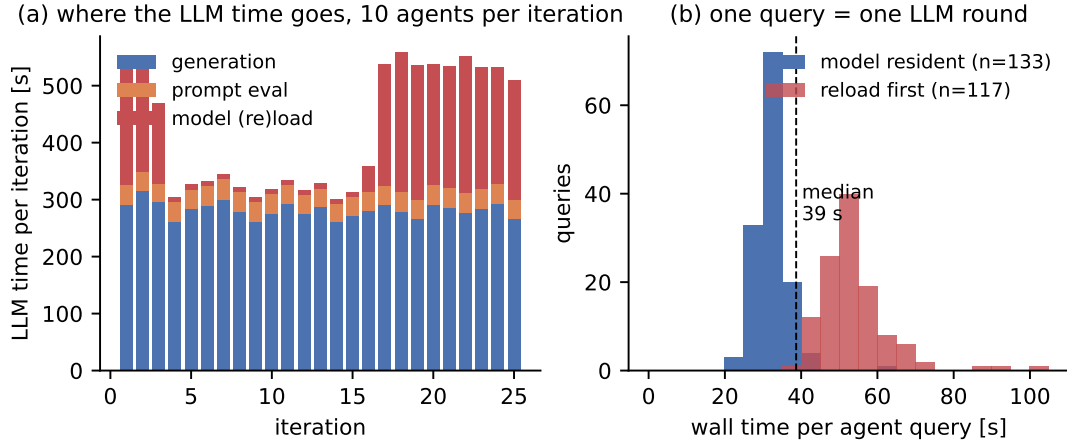


Figure 2: LLM time. (a) Per iteration, split into generation, prompt evaluation and model loading. (b) Wall time per query.

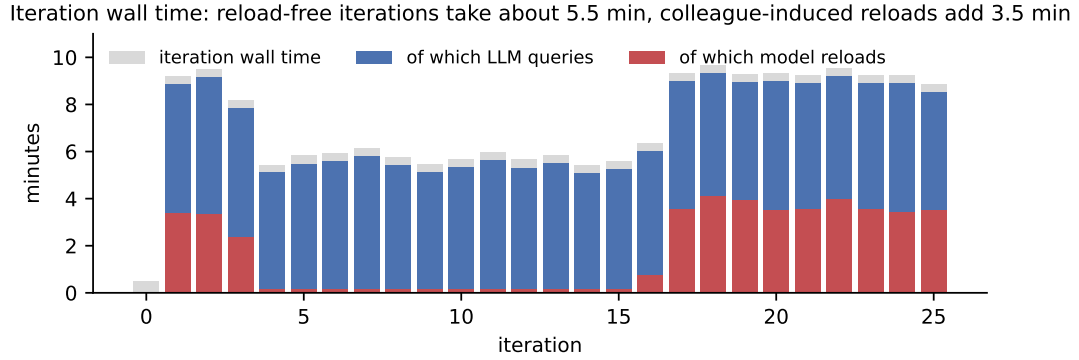


Figure 3: Iteration wall time. Grey: whole iteration; blue: the ten LLM queries; red: the part spent reloading the model after another user’s job.

4 What the model actually decides, and why

4.1 The decision is a rule

The model kept the day in 191 of 250 queries and changed at least one trip in 59 (42 agents); 102 of 113 changed trips became car trips (Fig. 6). Reading the decisions against the agent’s attributes and current modes removes the apparent variety (Fig. 7a): car owners switched 84 of their 86 bus trips and all 18 of their walk trips to car, and left all 270 car trips as they were; carless agents kept 105 of 112 bus trips (7 to walk) and 12 of 14 walk trips (2 to bus). Departure times were never touched. Whether the day changes is therefore decided by two facts, *owns a car* and *is not already driving*. Age and job appear in every trace but change nothing: the higher change rate of older car owners (Fig. 7b) is composition, older agents in this sample are more often on the bus. Re-selected agents confirm it: all 17 who had changed changed again (15 to car), 32 of 33 who had kept kept again. *Caveat:* the input that most often triggered the switch, the transfer count, was wrong (Section 4.2), so this rule describes the model’s reaction to a biased context, not necessarily its preferences.

4.2 Why some agents chose a slower mode

Figure 9b plots, for every bus trip of a car owner, the bus time against the car time the model was given; switches happen on both sides of the diagonal. Twelve conversations chose a mode slower than the current one for at least one trip. Read at the level of the whole day they split in two equal groups (Fig. 8a).

Six are not slower over the day. The agent accepts a slower morning drive because it brings the car to work for a much faster trip home. A 64-year-old woman took a 28-minute drive instead of a 15-minute bus, then drove home in 3 minutes instead of 16, and wrote out the sum herself: “it actually saves me time overall”. This is correct reasoning over the vehicle chain, not a disregard of the numbers.

Six make the day 13–34 minutes longer, knowingly. The traces state the times correctly and accept the loss to avoid transfers: “I’d rather sit comfortably in my car for an extra 15 minutes than shuffle through multiple bus/train connections” (53-year-old). Paying time to avoid changes is real behaviour; transport studies typically value one transfer at roughly 10–20 minutes of in-vehicle time.

But the transfer counts were wrong. The route comparison counted every leg of a transit trip as a bus leg,

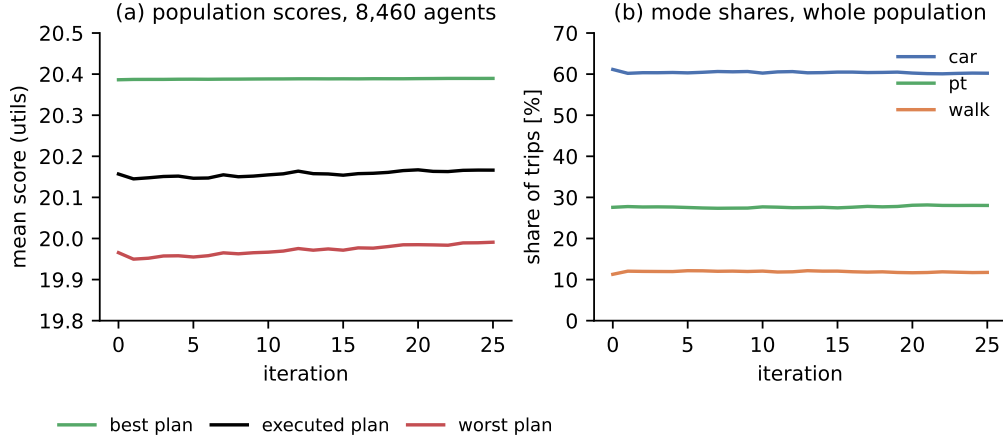


Figure 4: Ground health. (a) Mean score of the executed, best and worst plan. (b) Mode shares of all trips.

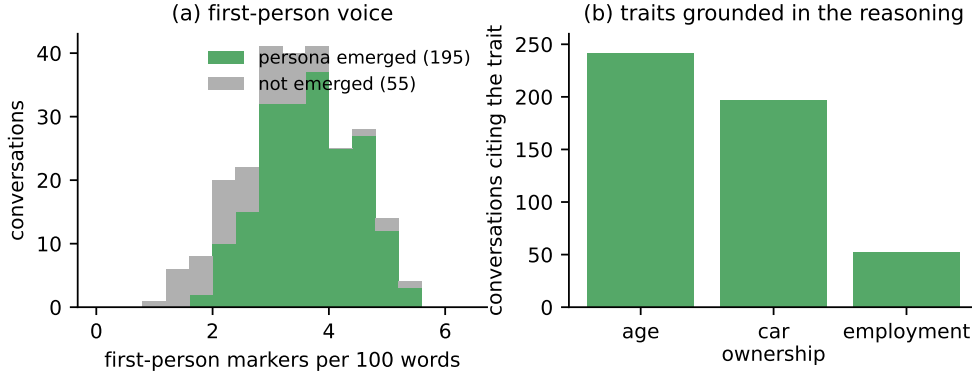


Figure 5: Persona signals in the reasoning traces. (a) First-person density. (b) Traits cited.

including the walk to the stop and the walk from it. A direct bus was therefore reported as 2 transfers, one change as 4, two changes as 6 (Fig. 8b); an option reported with 0 transfers was a walk-only fallback, identical to the walk option in 76 of 76 cases. In the final iteration 3,711 of 4,744 bus trips are direct, so the model was told that almost every bus ride needed two changes. The 53-year-old who drove 31 extra minutes to avoid “4 transfers each way” was avoiding a single change; the switch rate of 69 of 71 at “2 transfers” is a switch rate on direct buses. The persona’s valuation looks plausible; the fact it was valuing was not, and the model had no way to notice.

The car times were real. The 53-year-old was told 35 and 52 minutes by car and drove 37 and 49 minutes in the last iteration. With capacity scaled to 0.1 some corridors are congested in the evening peak, so the long car times are a property of the ground, not an error.

4.3 What the reasoning talks about

Figure 9a counts which considerations appear in the 250 traces (regular expressions over reasoning and answer). Travel time, the car-availability rule, age, transfers (inflated, Section 4.2), employment and comfort are near universal; walking distance and reliability appear in about 60%; weather in a third; parking in 22 traces; health in 11; money in 3. The persona almost never thinks about cost, which is the one term of MATSim’s utility function that could have argued for the bus; the context never mentioned cost either. Despite the instruction not to re-derive the numbers, 232 of 250 traces restate the given travel times in seconds and convert them to minutes before deciding; the median trace is 1,600 characters of reasoning followed by 800 characters of answer. Re-check loops (“Wait, ...”) are rare with the 3,072-token cap: two traces contain three or more, none hit the cap.

4.4 In the model’s words

4.5 Did the plans survive scoring?

For the 42 agents whose day the model changed (Fig. 10): 16 no longer hold a plan with the LLM’s modes (it scored worst of six and was removed), 26 still hold one, and for 15 it is the best-scoring and executed plan. Among the 24 agents holding both kinds of plan the score difference is tiny: median +0.05 utils, 20 of 24 within ± 0.5 . Under capacity 0.1 the car and bus plans of these commuters score about the same, so MATSim’s selection is close to a coin toss and the verdict is neither confirmation nor rejection.

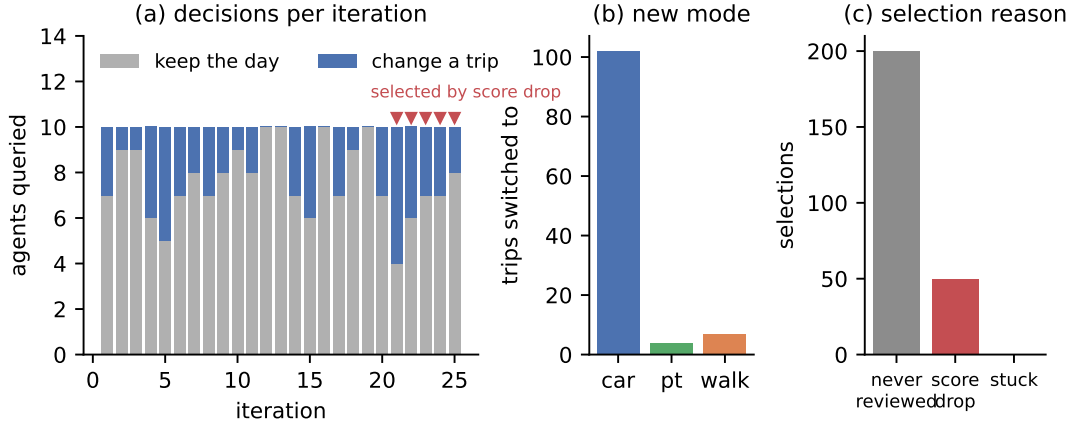


Figure 6: Decisions. (a) Keep versus change per iteration; red markers: iterations with score-drop selection. (b) New mode of the 113 changed trips. (c) Selection reasons.

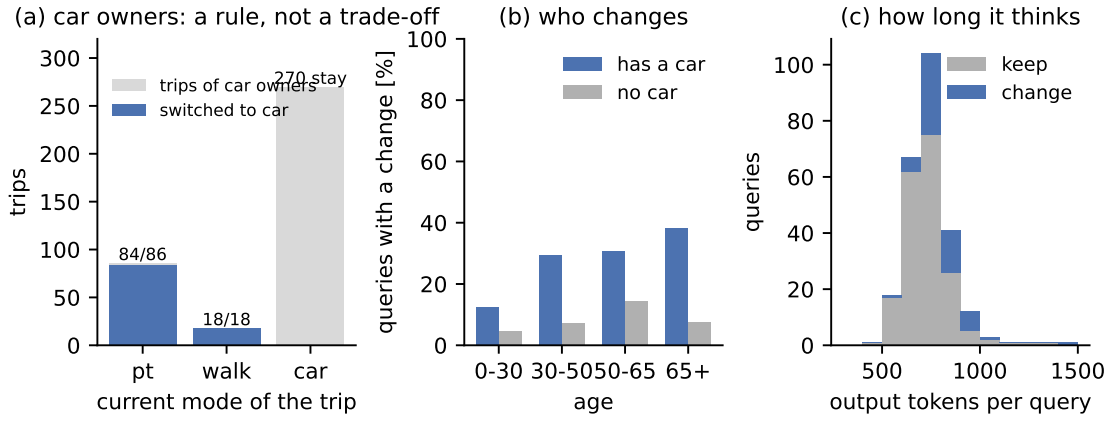


Figure 7: Who changes. (a) Car owners’ trips by current mode and whether they were switched to car. (b) Share of queries with a change, by age and car access. (c) Output tokens per query.

5 What was not achieved

No behavioural result. Three design choices made the run flat: the LLM only proposes and MATSim’s utility function judges, so a preference the utility does not share (avoiding transfers) is scored away; the ground was already at equilibrium, so there was nothing to improve and the score-drop trigger selected noise (0.2 utils); and 10 agents per iteration is 0.1 % of the population, too little for any network effect. The run characterises the instrument, not the phenomenon.

No individuality. The persona template gives the model three facts (age, employment, car access). The decision uses one of them. Everything else in the trace is narrative around a fixed rule. A population of such agents would be homogeneous within each car-access class, which is not what a persona approach is for. How much of this homogeneity is the model and how much is the inflated transfer count cannot be separated in this run.

No learning. The experience handler wrote 20,799 per-trip facts for the panel to the vector store during the run (“On day 6, the car trip from link 56_4 to link 53_1 took about 5 minutes 44 seconds during the afternoon peak”). None of them reached a prompt: 0 of 250 requests contained retrieved context. The retrieval code that handles a list of messages builds the enriched message and then discards it; whether the single-message path retrieved nothing for another reason is still to be checked. The persona therefore never saw yesterday’s outcome or its own past decisions, which is why re-selected agents simply repeat themselves.

No verdict from scoring. Car and bus plans score alike for these agents, so the run cannot say whether the persona’s rule is “right” in MATSim’s terms. A rule-based control on the same panel is needed before the 15/42 survival can be interpreted.

Tooling gaps. The Python analysis suite reports 0 % success and 0 mode changes for this run because it looks for the old full-plan tool; the analyses above were done with two ad-hoc scripts kept next to this report. A departure-time change was never proposed, although the tool allows it.

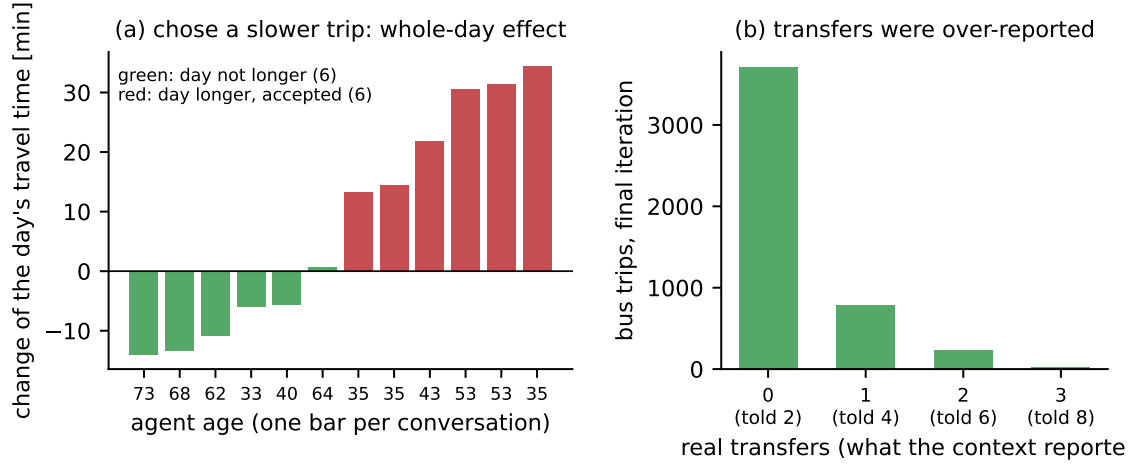


Figure 8: (a) The twelve conversations that chose a slower mode for some trip: change of the day’s travel time, as given to the model. (b) Transit trips of the final iteration by real number of transfers, with the number the context reported for the same trip.

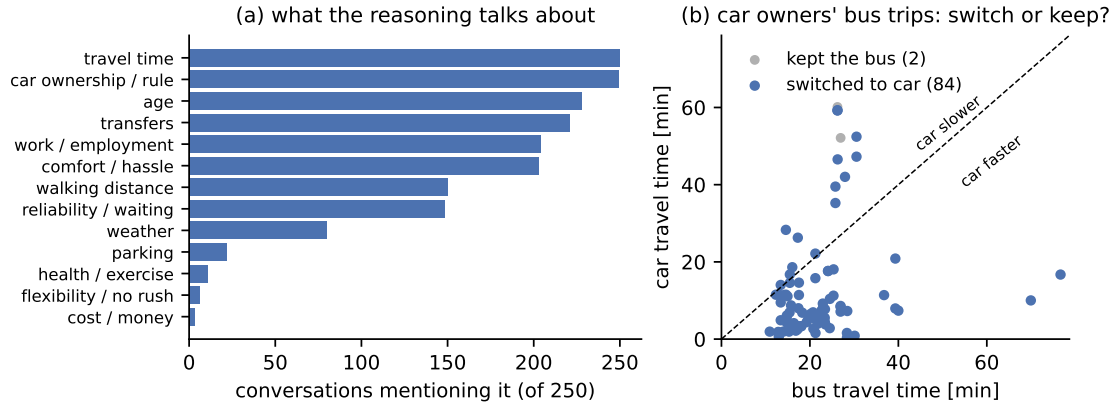


Figure 9: (a) Considerations mentioned in the reasoning traces. (b) Car owners’ bus trips: bus time versus car time as given to the model, coloured by decision. The diagonal is equal time.

6 Issues found

Transfer counts over-reported (open). Described in Section 4.2: the route comparison counts access and egress walks as bus legs, so every transit option in every prompt carried two phantom transfers. It affects all runs that used the comparison tool, including the earlier tool-calling runs. The fix is to count only the bus legs. The wider lesson matches the first bug below: the model cannot verify the facts it is given, so every number in the context has to be checked against what the simulation actually produces.

A context bug that looked like a result. The first attempt at this run was stopped after two iterations: 23 of 24 decisions were “keep”. The context listed the modes available *at each place under the current plan*; a car owner on the bus has the car at home, so the list said “from work: walk, bus, passenger” and the return trip had no car option. With the rule “the car is wherever you last parked it”, the model concluded it could not drive home and did not drive out either. The fix presents vehicles at the level of the whole day, with the car option on every trip that an unbroken chain can bring it to, marked “requires driving trip 1 as well”; the decision check follows the same chain. Same seed, same first ten agents: 0 changes before, 3 after (Fig. 11). General lesson: a fact that is conditional on the plan being changed must be stated as a conditional, or the model reads it as a constraint.

53, has a car, switched to car	“Looking at this plan, I’m supposed to take public transport both ways with 4 transfers each time. That’s a lot of walking around and waiting for connections, especially at my age. Four transfers means getting on and off multiple times, which is tiring.”
76, no car, kept the bus	“Looking at my day, I’m heading out from home at 11:17 to a secondary activity, staying until 12:32, then coming back home. That timing feels fine for me, a nice midday outing. I don’t have a car or bike available, so my choices are really just public transport or walking.”
43, no car, switched to walk	“I’m struck by how many transfers the public transport trips involve. Trip 1 has 6 transfers and trip 2 has 8 transfers! That’s exhausting, getting on and off buses or trains that many times, waiting at stops, dealing with connections.”
80, no car, thinking trace	“PT takes about 35 mins and involves transfers (2 transfers). Wait, the route options say transfers: 2 for PT. That means getting on/off multiple times, which can be tiring for an elderly person. But walking 45 minutes straight might also be tough. Actually, PT is faster and less physically demanding than walking 2.2 km.”

Table 2: Representative answers. The voice is consistent and plausible. Transfer counts quoted by the model are the inflated ones: “4 transfers” is one change, “6 and 8” are two and three, “2 transfers” is a direct bus.

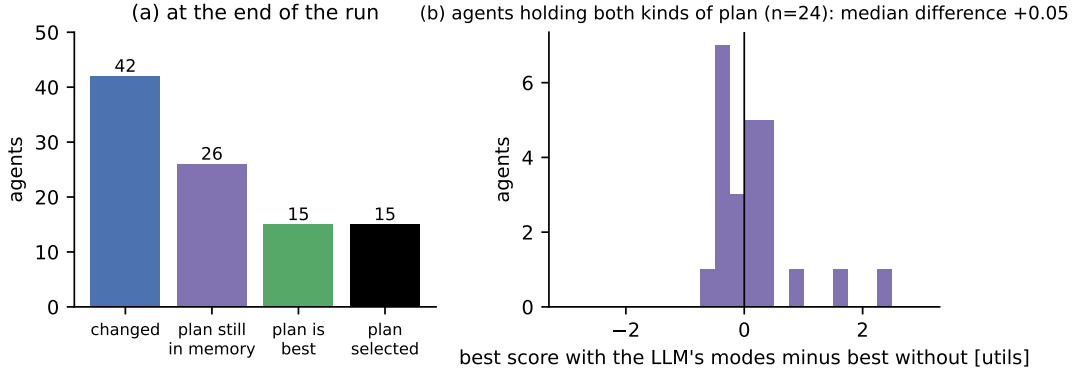


Figure 10: Fate of the 42 changed days at the end of the run. (a) Plans still in memory, best-scoring, executed. (b) Best score with the LLM’s modes minus best without, for agents holding both.

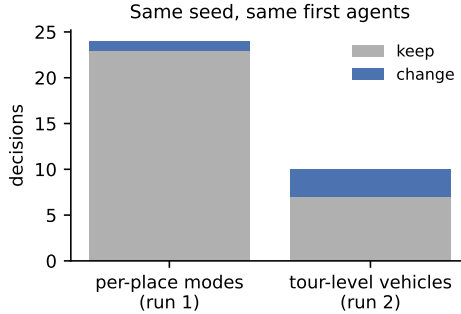


Figure 11: Same seed and first agents, before and after the context fix.

Stricter physics for the LLM than for MATSim. MATSim’s rule-based strategies gave one panel agent (9632_1) a car-out, bus-back day, which leaves the car at work overnight; the vehicle-chain check forbids exactly that for the LLM. The two kinds of strategy do not compete under the same rules.

Shared GPU. A colleague’s job using the same model name at another context size forced a reload in 117 of 250 queries, a quarter of all LLM time (Fig. 3). Our eviction script evicted by model name and could have unloaded their instance; it now checks the context size. **Innovation** was never switched off (the runner leaves MATSim’s default), so the LLM ran in all 25 iterations; harmless here, to be decided for future runs. **The laptop** ran the MATSim side and had to stay open for three hours; moving it to the server is the remaining engineering step.

7 Next

1. **Correct the inputs first.** Count only bus legs as transfers, then audit every number in the context against the executed trips of the same iteration, and repeat this run’s decision analysis on corrected inputs before building on it.
2. **Final word for panel agents, with guards.** Panel agents keep only the LLM’s plan; MATSim computes the consequences. The model still cannot judge feasibility, so the guards stay: the vehicle-chain check,

MATSim routing, and a fallback to the previous plan if the executed day is stuck or infeasible. Then the population moves by what the personas decide.

3. **Experience in the loop.** Yesterday’s realised travel times, delays and score change, plus the agent’s own past decisions and reasons, injected as a short structured block (not raw transcripts), stored per agent in the vector store already in place. This is the mechanism for consistency *and* for change.
4. **A shock, a cohort, a control.** Start from the warmed ground, remove a bus line or halve a corridor’s capacity, decide a cohort of 100–200 agents by the LLM in the same iteration, with a rule-based cohort as control, and compare adaptation.
5. **Speed.** 32s per query is 745 generated tokens at 26 tokens/s. The levers, in order of gain: shorter or no reasoning on most queries (a decision alone is about 50 tokens, roughly 5s), our own model instance with parallel decoding on the server, and the 9B model for screening; the 27B with reasoning kept for a sample of queries so the persona metrics remain measurable.